MINISTRY OF EDUCATION
IMAM ABDULRAHMAN BIN FAISAL UNIVERSITY
COLLEGE OF COMPUTER SCIENCE & INFORMATION TECHNOLOGY

وزارة التعليم
جامعة الإمام عبد الرحمن بن فيصل
كلية علوم الحاسب وتقنية المعلومات

جامعة الإمام عبد الرحمن بن فيصل
IMAM ABDULRAHMAN BIN FAISAL UNIVERSITY

CS 321: Object Oriented Programming (2)

Final Project Report

Artistry shop

Figure 1: Project logo

GROUP NUMBER (IN BLACKBOARD): 3

SECTION NUMBER: 11

MEMBERS:

	Name	ID	Role Participated in	Sign
1	(Leader): Sara Almary	[redacted]	GUI, Coding	
2	Fatimah Al Yousef	[redacted]	GUI, Coding	
3	Bayan Almadan	[redacted]	Database	
4	Fatimah AlMarhoon	[redacted]	Exception Handling	
5	Zainab Al-Marzouq	[redacted]	Database	

Table 1: Names and ID

INSTRUCTOR NAME: Alaa Alahmadi

DATE OF SUBMISSION: 20 May 2024

DIFFICULTIES & HOW PROBLEMS WERE SOLVED:

We started our project from the beginning of the semester, yet we passed through a lot of difficulties. The biggest problem was how to connect the database to NetBeans, we solved this problem after we studied it in the lab section, but some of our team members -although followed the steps written in the lab instructions- couldn't connect the database with NetBeans. So, we searched for another solution and asked those around us.

Another problem was how to use the database in the code by adding to it or displaying it, but it was solved after a bit of research.

How to update employee information was another problem, but it was solved after finding a tiny segment of the code that was causing the error and correcting it.

CHECKLIST OF REQUIREMENTS FULFILLED IN THE PROJECT:

Export and import data from files.
Multi-part forms, logo, navigation, login control.
Database design, constraints, DDL, DML, read queries, login facility, database connection.
User exception handling.
Updated ER, Mapping, Diagram, tables.

HOW WERE THE TASKS DISTRIBUTED EQUALLY BETWEEN MEMBERS?

We start by working together on the basic things what is the program is about and the idea of it, after that we write a list of the tasks that we should do,
then divide them between us:-
Sara: Admin Event, Admin menu, Admin paintings, Admin supplies, database connection, Employee update, Homepage, Items, add picture, PDF receipt, User shop.
Fatima Al Yousef: Art shop, cart, database connection, Admin paintings, Admin supplies, Customer information, Employee update, Items, add pictures, User shop.
Bayan: create the database, fields, tables, relationship. User Login, User sign up, Admin Login, database connection, Evaluation.
Fatima Almarhoon: Try and catch for all project (Admin Event, Admin menu, Admin paintings, Admin supplies, database Connection, Employee update, Homepage, Art shop, cart, Items, add picture, PDF receipt, User Login, User sign up, Admin Login, User shop).
Zainab: create the database, fields, tables, relationship, database connection, Employee update, Admin Event.
After we finish work individually, we start to collect the work together and try the program if there are any mistakes in run-time error or logical error, then we make share filed and draft the report in it, same thing for project presentation.

ER:

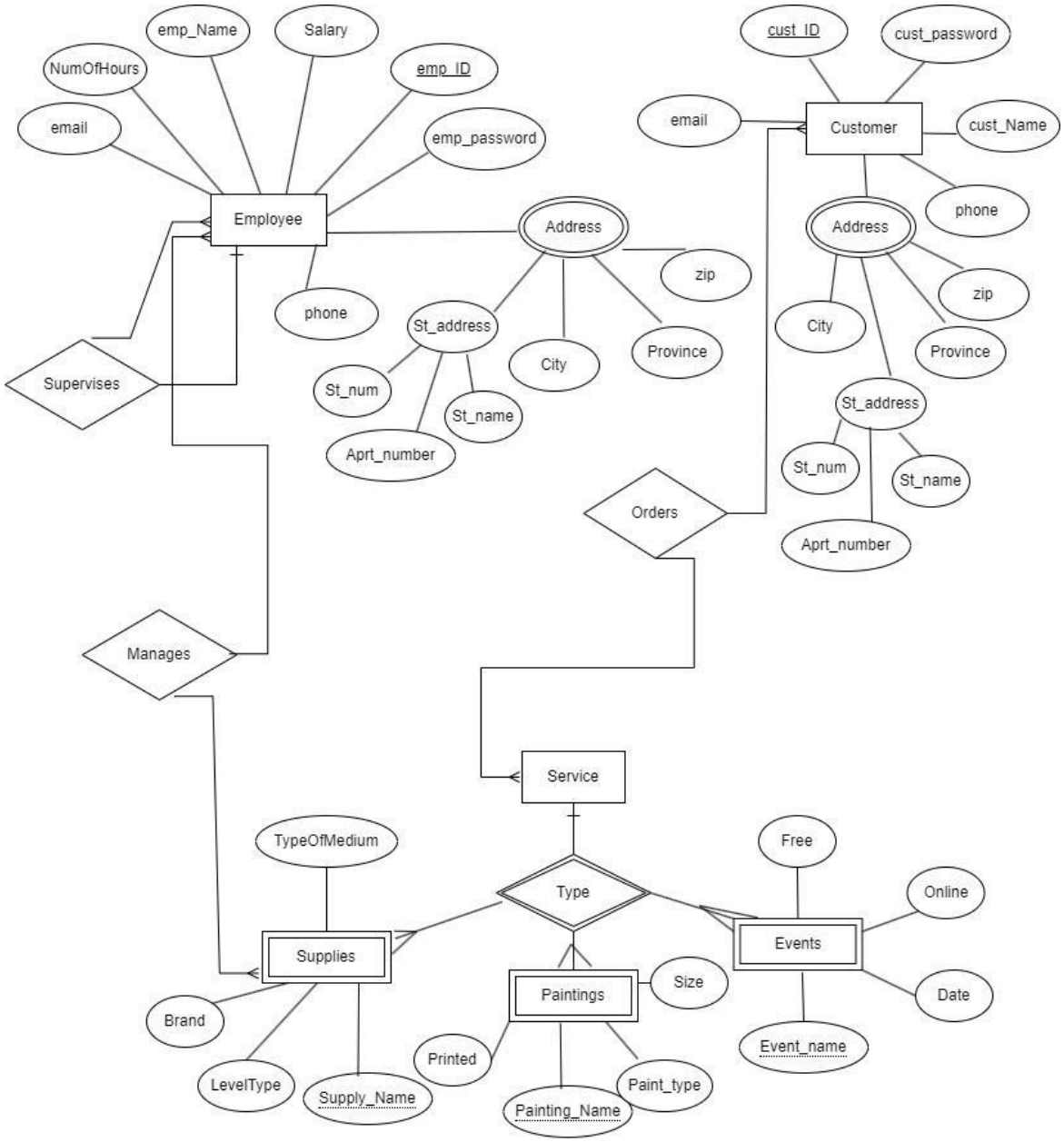


Figure 2: ER Diagram of the application

MAPPING of ER diagram into Database Schema:

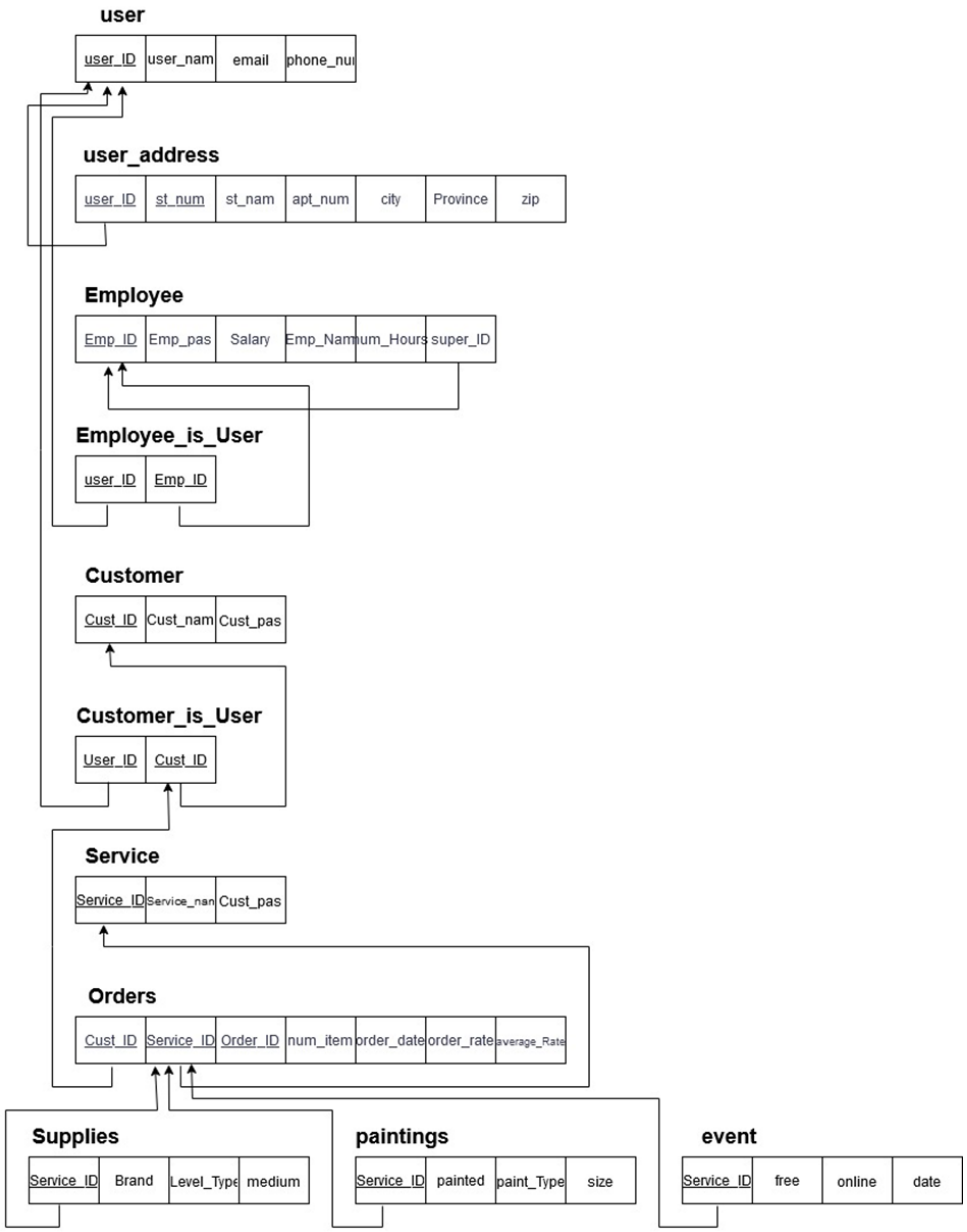


Figure 3: Relational Schema of the application

Mapping tables:

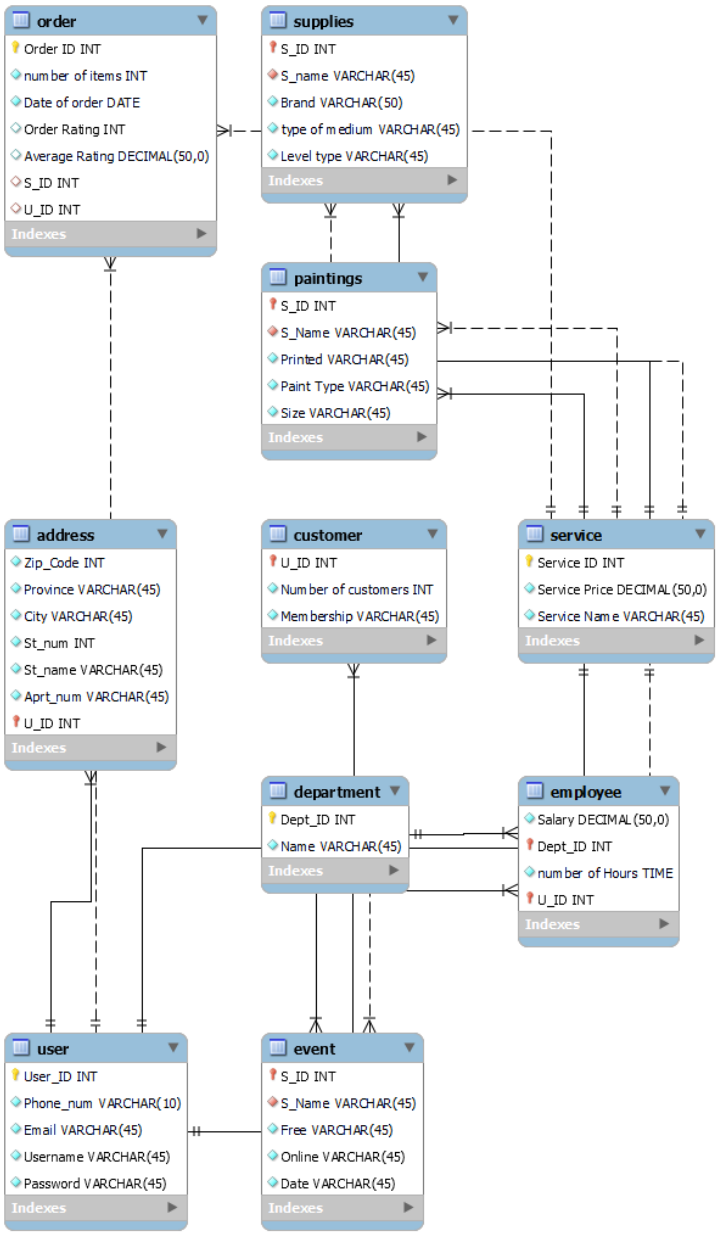


Figure 4: Mapping Tables of the application

Tables in database:

Address table

Zip_Code	Province	City	St_num	St_name	Aprt_num	U_ID
5668	Eastern Region	Al Qatif	3	uhud	3595	1
7974	Eastern Region	Al Qatif	4	uhud	3856	2
8985	Eastern Region	Al Qatif	2	Abu tahir	7585	3
3478	Eastern Region	Al Qatif	8	Al quds	3896	4
8979	Eastern Region	Al Qatif	5	Abu tahir	5687	5
1997	Eastern Region	Al Qatif	7	Yaqut	6438	4
6393	Eastern Region	Al Qatif	5	Al jazeera	2479	5
5466	Eastern Region	Al Qatif	3	uhud	4792	6
5324	Eastern Region	Al Qatif	9	Alquds	6973	7
5355	Eastern Region	Al Qatif	8	Yaqut	1679	8
8538	Eastern Region	Al Qatif	2	Bader	9763	9
3367	Eastern Region	Al Qatif	1	uhud	3685	10
NULL	NULL	NULL	NULL	NULL	NULL	NULL

Figure 5: Address table in database

Customer table

U_ID	Number of customers	Membership
1	5	Yes
2	6	Yes
3	4	No
4	16	Yes
5	5	No
6	4	Yes
7	6	Yes
8	4	No
9	7	Yes
10	5	No
NULL	NULL	NULL

Figure 6: Customer table in database

Department table

Dept_ID	Name
111	Dammam
222	Al Qatif
333	Riyadh
444	Khobar
555	Jubail
666	Khafji
777	Nairiyah
888	Ras Tanura
999	Dammam2
1000	Riyadh2
	NULL

Figure 7: Department table in database

Employee table

Salary	Dept_ID	number of Hours	U_ID
4000	111	06:00:00	1
4000	888	06:00:00	2
4500	333	08:00:00	3
4000	999	06:00:00	4
4000	222	06:00:00	5
4000	333	06:00:00	6
4000	444	06:00:00	7
4000	555	06:00:00	8
4000	666	06:00:00	9
4000	777	06:00:00	10
NULL	NULL	NULL	NULL

Figure 8: Employee table in database

Event table

S_ID	S_Name	Free	Online	Date
102	Student Showcase	No	NO	Every Two Weeks
103	Group Art Wall	Yes	NO	1st day on every month
NULL	NULL	NULL	NULL	NULL

Figure 9: Event table in database

Order table

Order ID	number of items	Date of order	Order Rating	Average Rating	S_ID	U_ID
1818	6	2024-03-29	4	17	102	4
2662	3	2024-04-26	7	26	77	10
4333	1	2024-04-13	8	30	46	8
4343	5	2024-04-15	5	22	19	9
4444	10	2024-02-14	5	20	2	1
5555	9	2024-03-15	5	20	201	2
6666	4	2024-04-04	3	13	4	5
7444	6	2024-05-01	5	20	102	7
8989	2	2024-05-06	7	27	4	3
9090	12	2024-03-20	5	19	203	3
NULL	NULL	NULL	NULL	NULL	NULL	NULL

Figure 10: Order table in database

Paintings table

S_ID	S_Name	Printed	Paint Type	Size
2	The Starry Night	No	oil	20x40
4	Irisis In Monets Garden	Yes	ink	18x24
23	skatch book	Yes	None	30x15
77	coloring book	yes	ink	20x20
NULL	NULL	NULL	NULL	NULL

Figure 11: Paintings table in database

Service table

Service ID	Service Price	Service Name
2	200	The Starry Night
4	160	Irisis In Monets Garden
19	6	eraser
23	50	skatch book
46	25	Acrylic colours
77	30	coloring book
102	25	Student Showcase
103	0	Group Art Wall
201	20	Paint Brush
203	100	Brush set
NULL	NULL	NULL

Figure 12: Service table in database

Supplies table

S_ID	S_name	Brand	type of medium	Level type
201	Paint Brush	Winsor & Newton	Acrylic	Beginner
203	Brush set	Derwent	Oil	Professional
23	eraser	rubber	paper	All
46	Acrylic colours	Faber-Castell	acrylic	Beginner
NULL	NULL	NULL	NULL	NULL

Figure 13: Supplies table in database

User table

User_ID	Phone_num	Email	Username	Password
1	0510024356	Fatima_Y@hotmail.com	Fatimah Al Yousef	Fati2134
2	0587637465	sara@gmail.com	Sara Almary	Luka999
3	0567359846	Bayan@outlook.com	Bayan Almadan	Bayan2032
4	0589376847	zainab@yahoo.com	Zainab Al-Marzouq	Zainab6789
5	0548749873	fatemah@gmail.com	Fatemah Al Marhoon	Moon9898
6	0563894799	dema@gmail.com	Dema Salih	Dema3232
7	0528395643	Lana@yahoo.com	Lana Almajed	Lana2200
8	0578927489	Rayan@outlook.com	Rayan Omar	Rayan5950
9	0568295683	Maryam@hotmail.com	Maryam Alnoor	Maryam1000
10	0578844567	Mona@gmail.com	Mana Jassum	Mona1999
NULL	NULL	NULL	NULL	NULL

Figure 14: User table in database

Run the program:

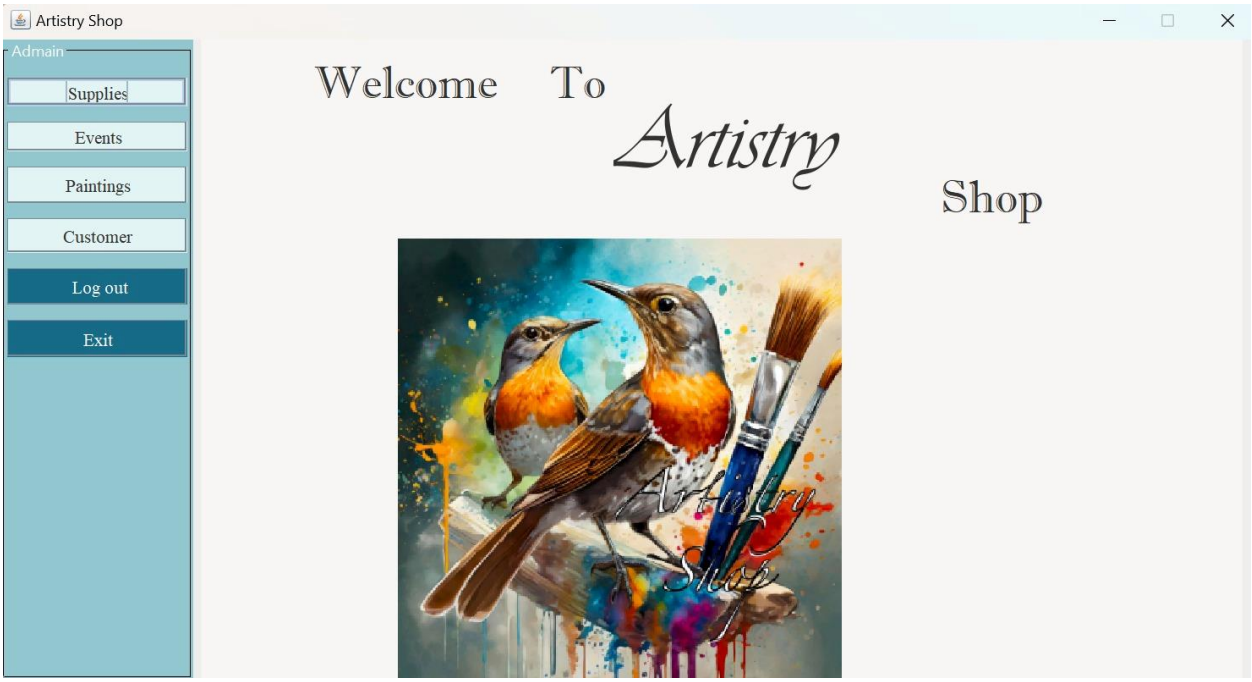


Figure 15: welcome to art shop

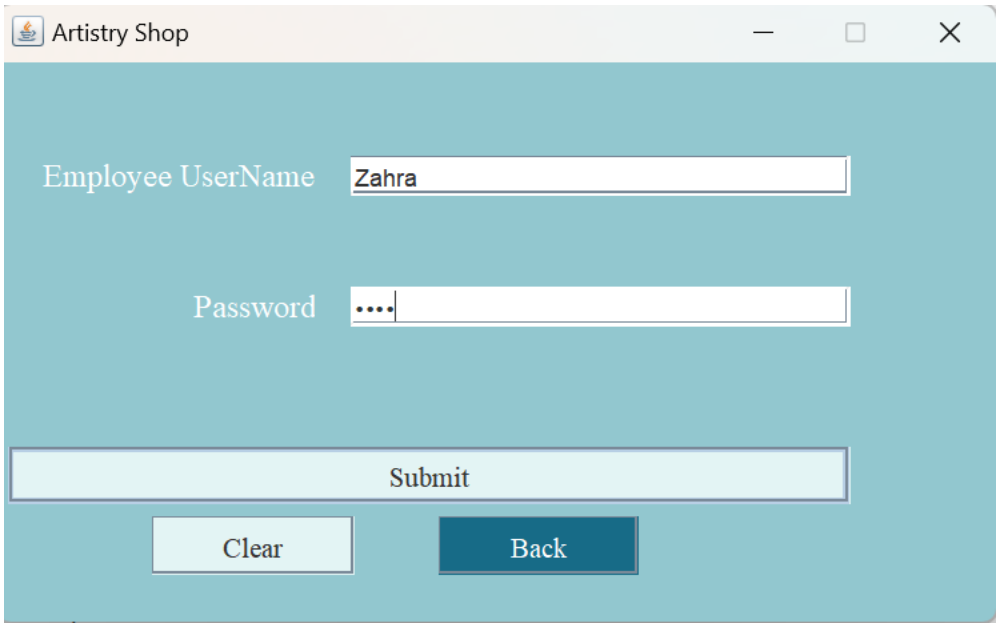


Figure 16: manager log in

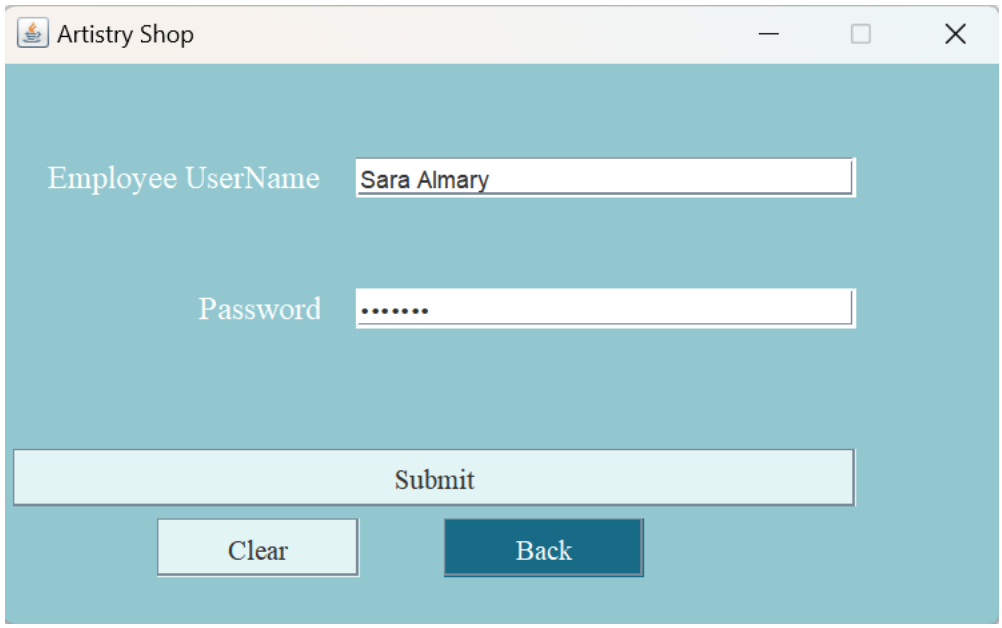


Figure 17: employee log in

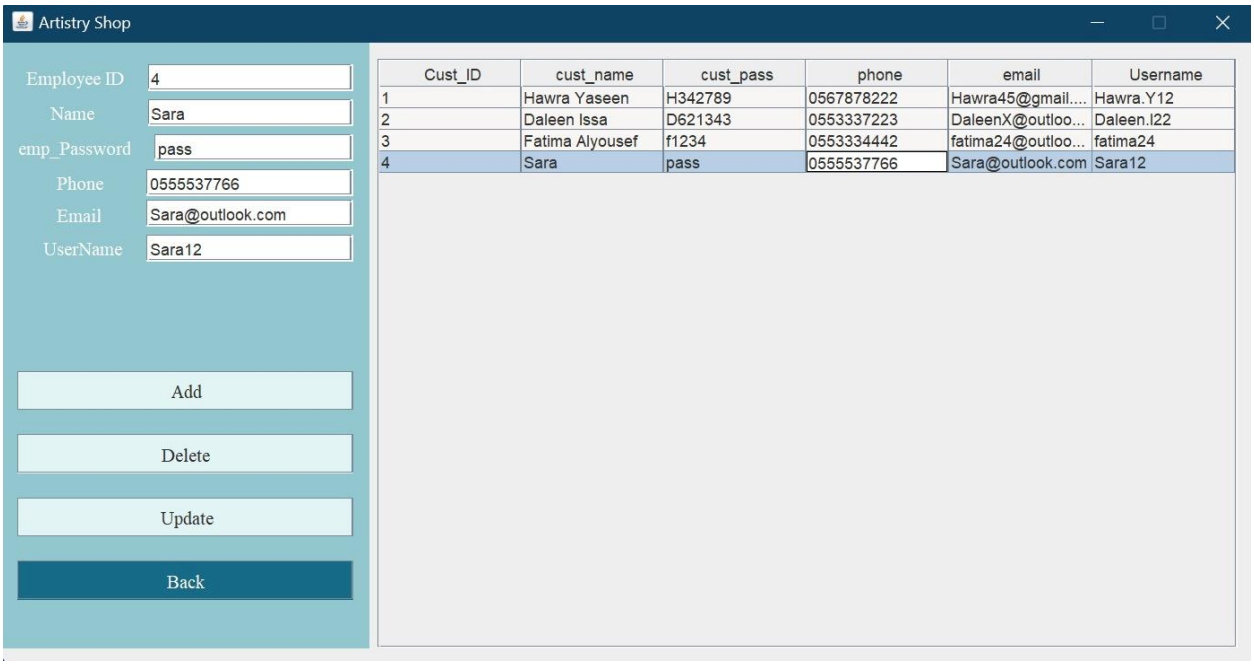
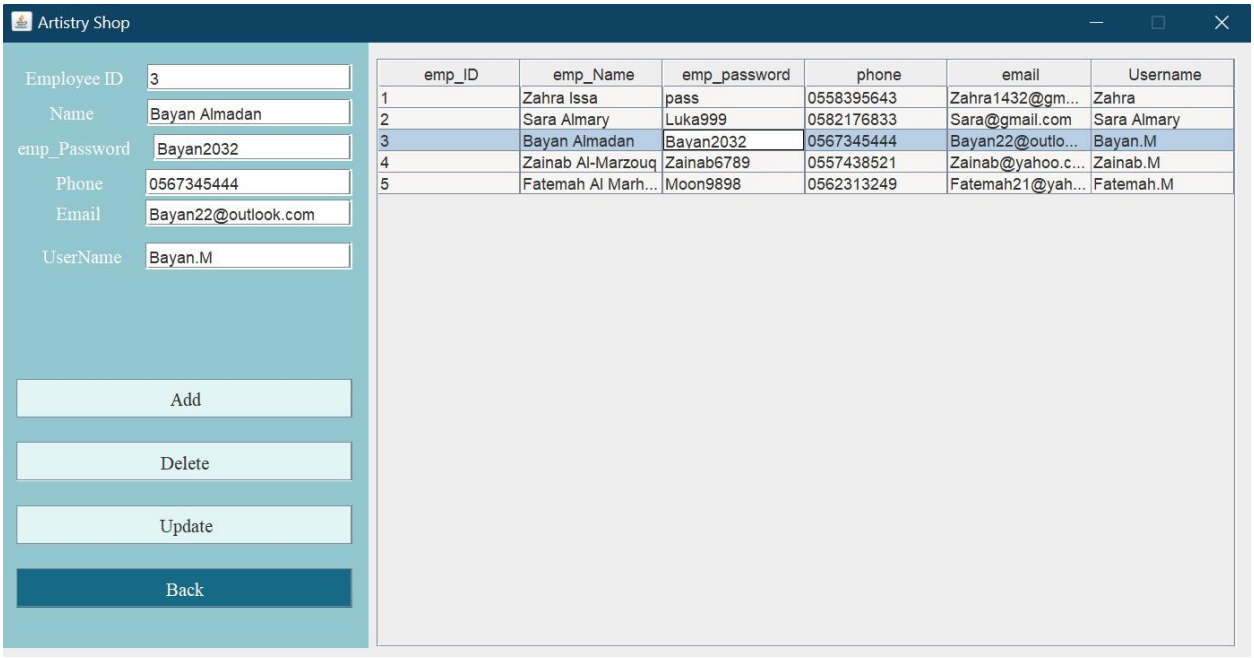


Figure 18: employee costume



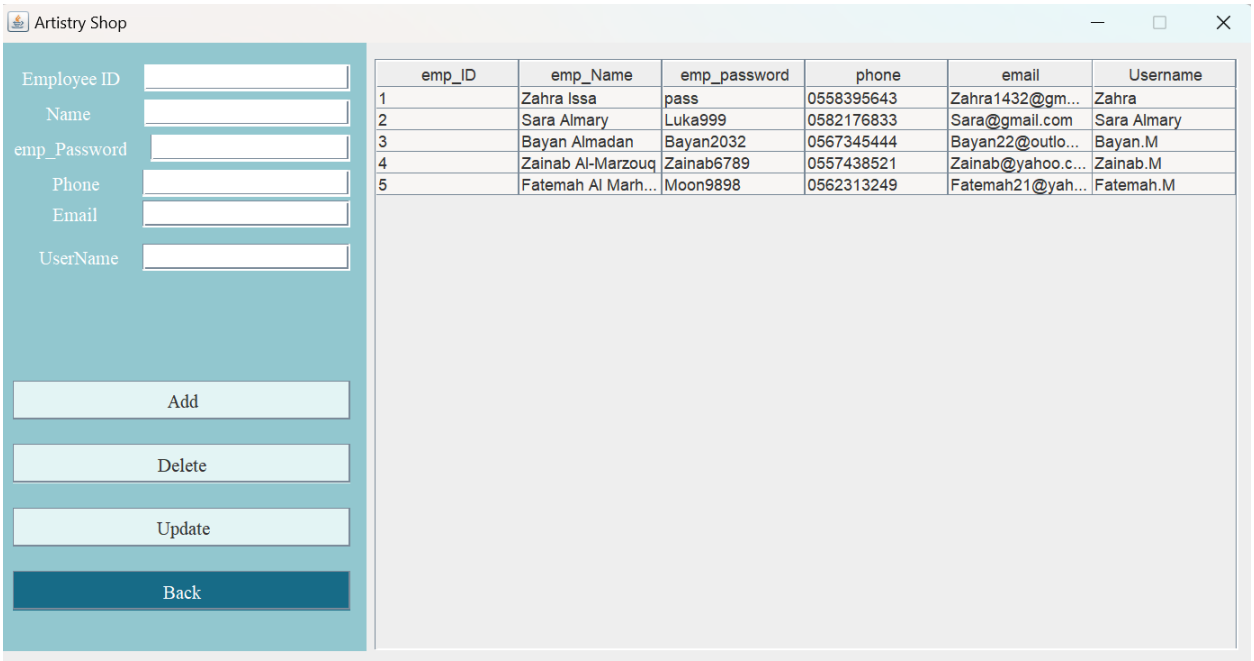


Figure 19-20: employee update

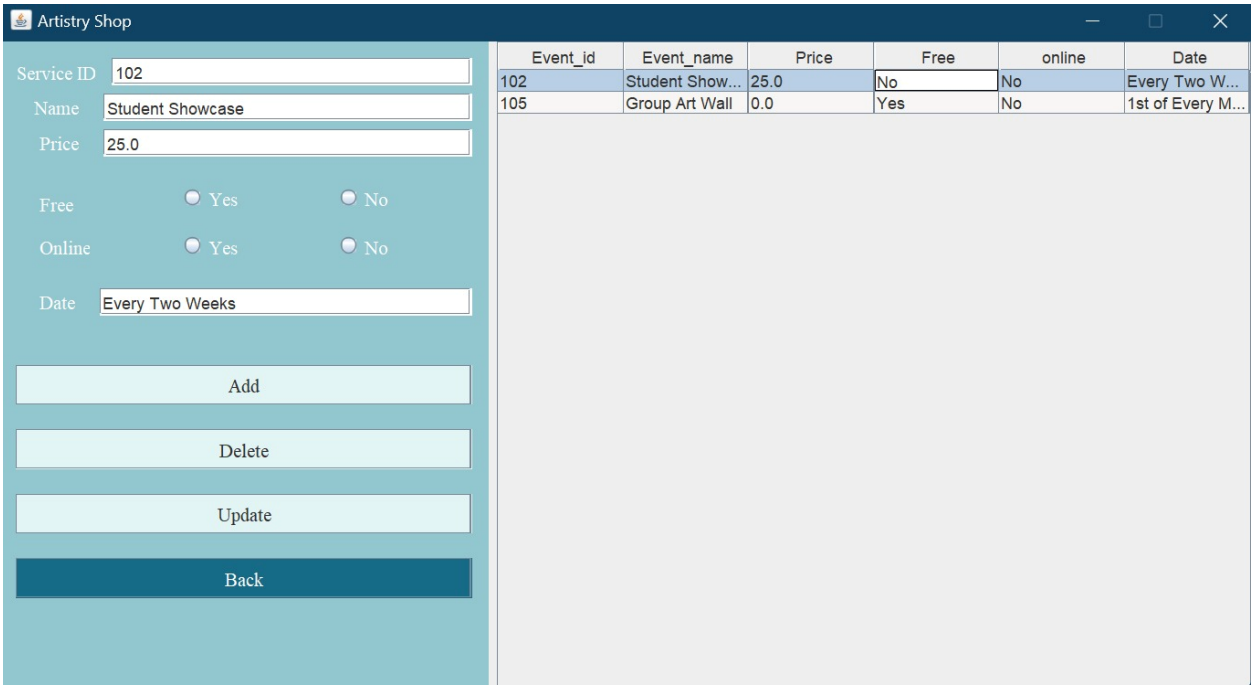


Figure 21: employee event

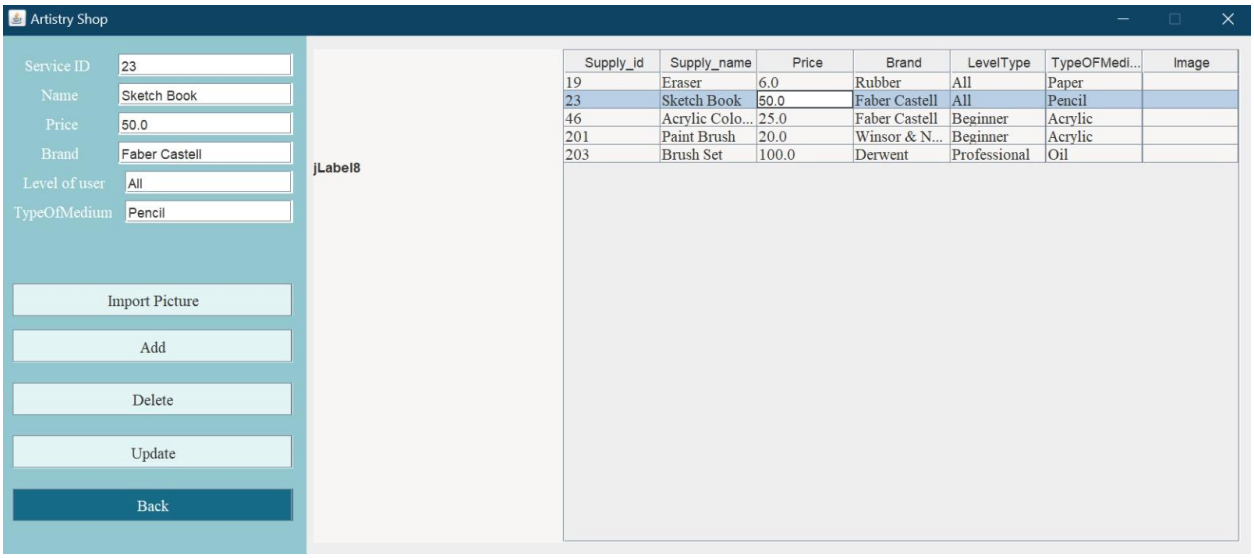


Figure 22: supplies

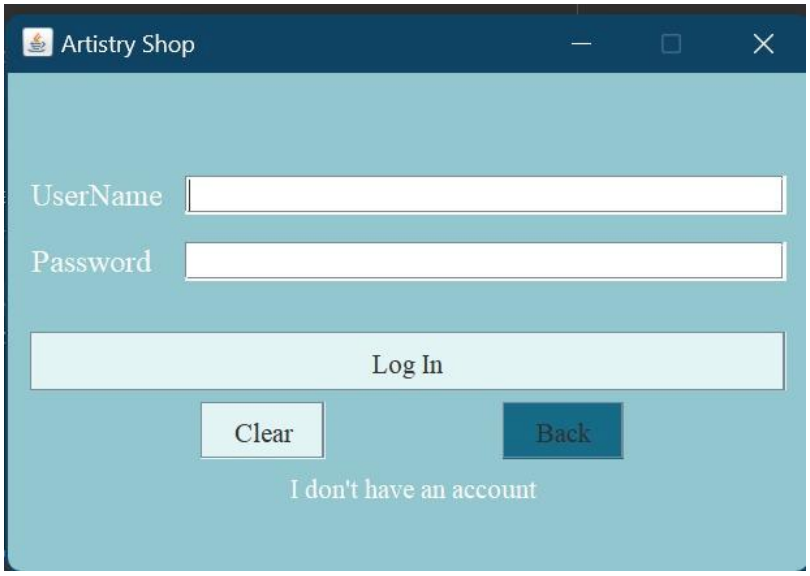


Figure 23: customer log in

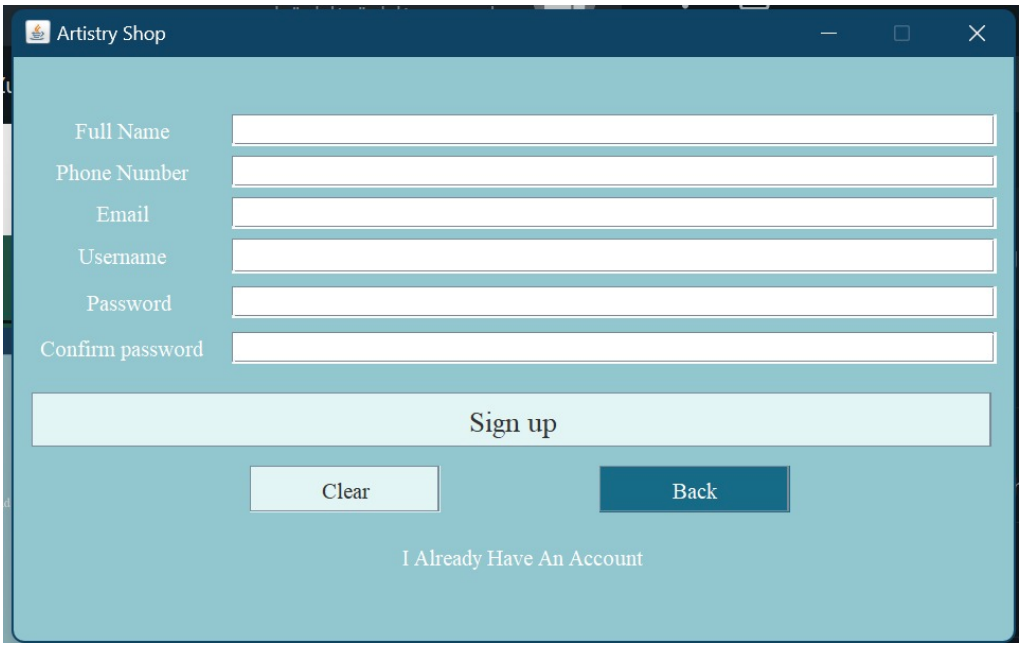


Figure 24: costumer sign up

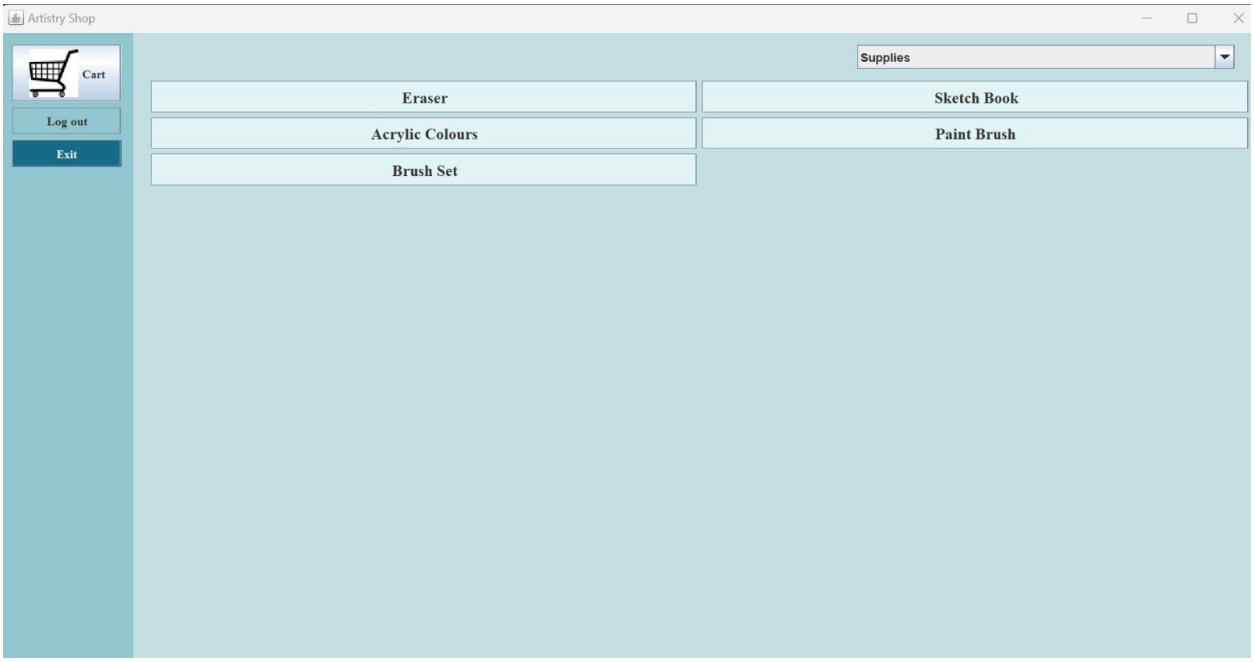


Figure 25: costumer supply

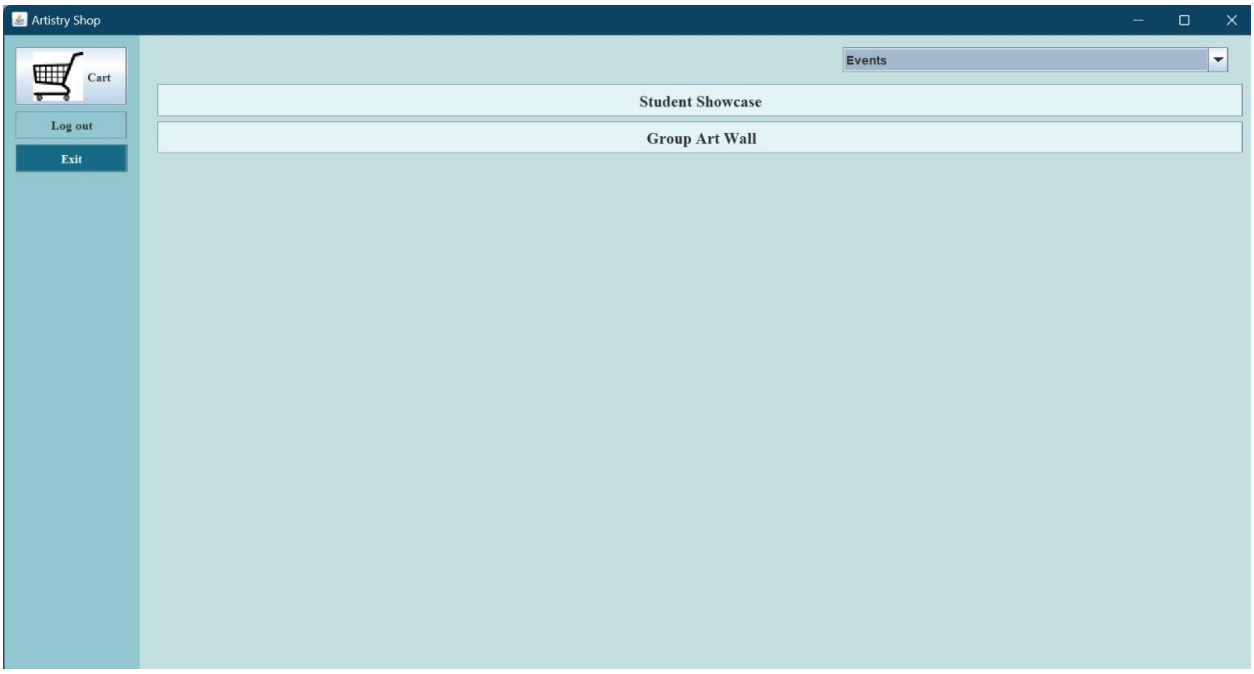


Figure 26: costumer event

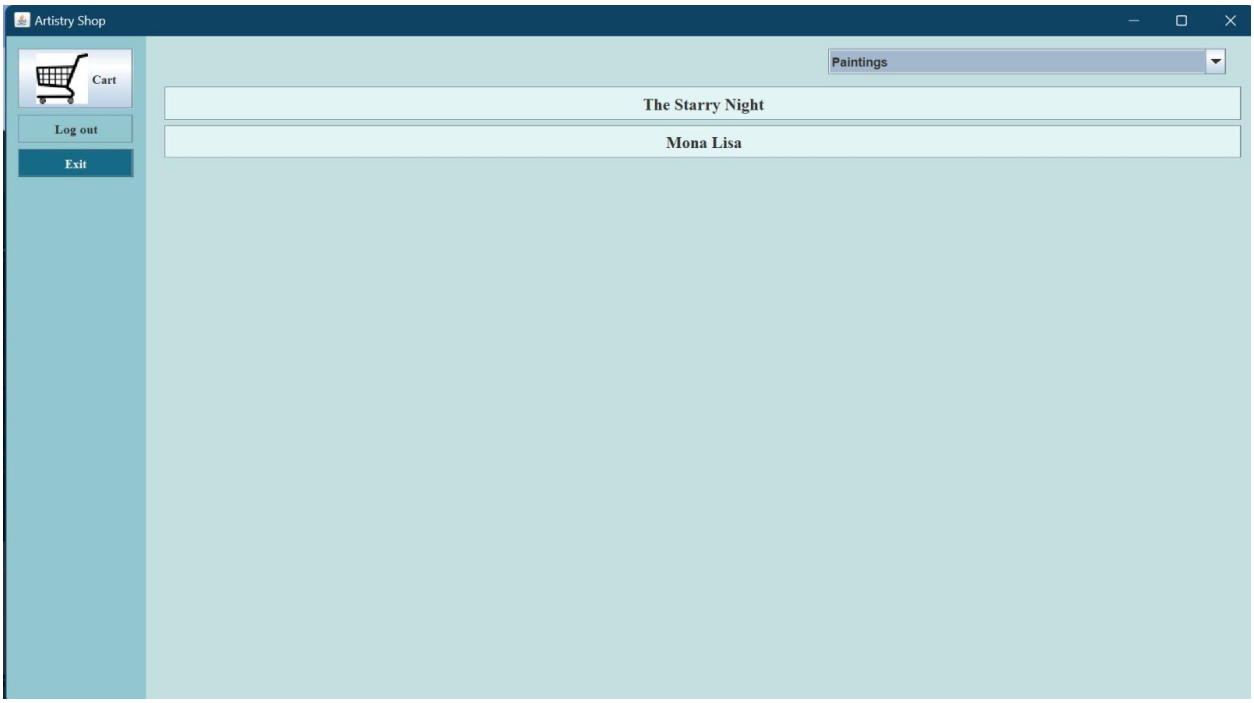


Figure 27: cart paintings

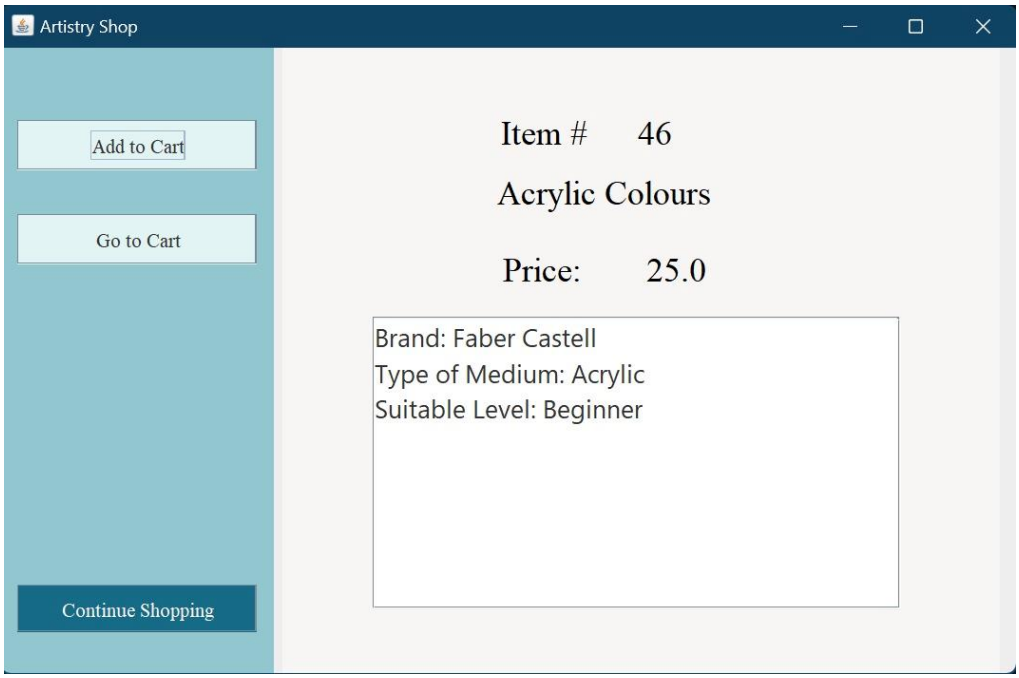


Figure 28: item frame

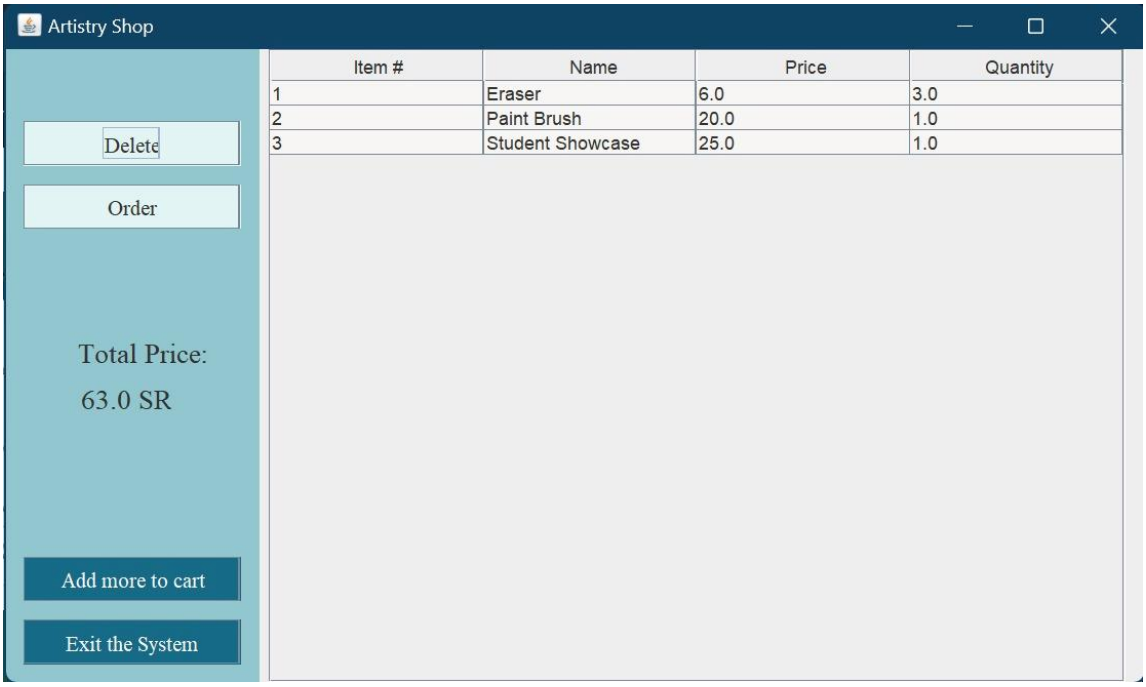


Figure 29: the cart

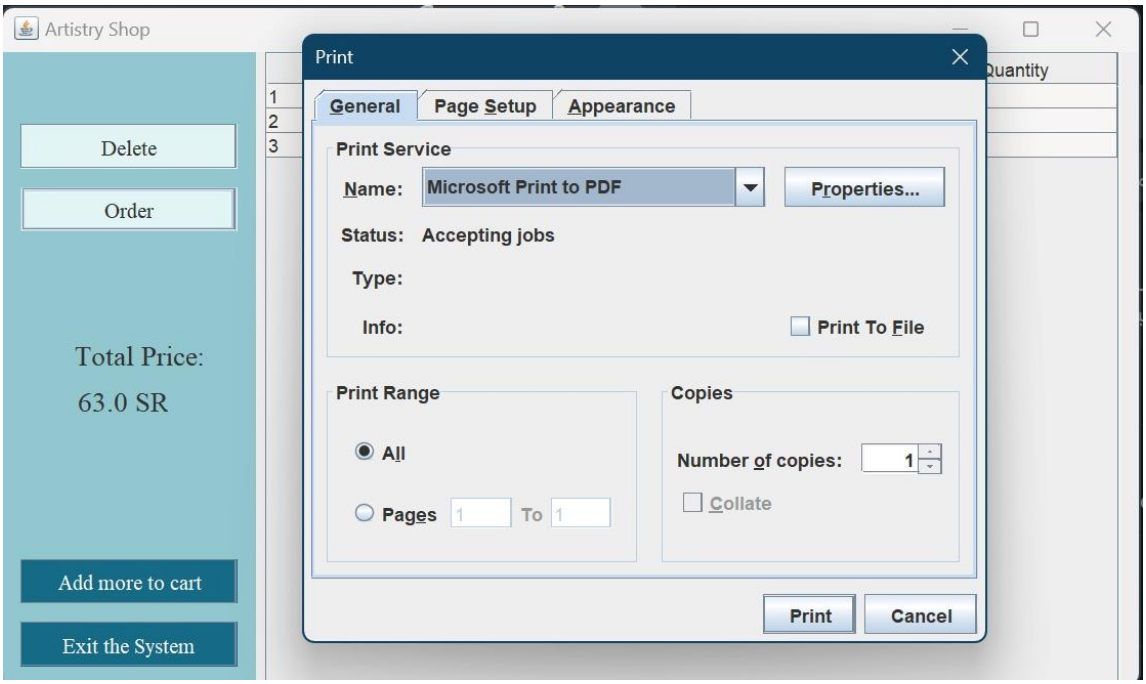


Figure 30: order

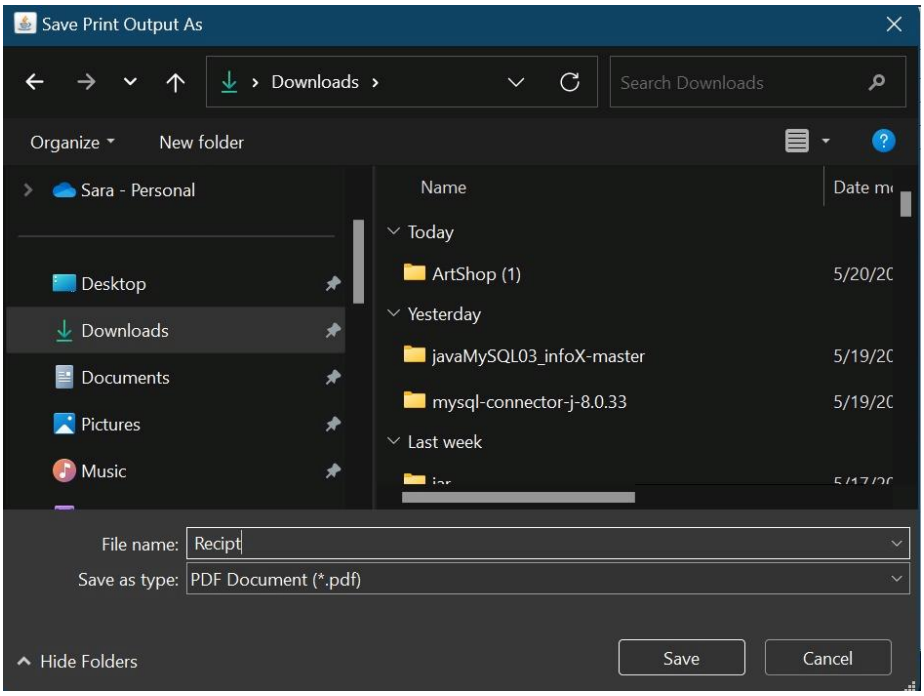


Figure 31: save to a file

Receipt

Item #	Name	Price	Quantity
1	Eraser	6.0	3.0
2	Paint Brush	20.0	1.0
3	Student Showcase	25.0	1.0

Total Amount: 63.0 SR

Figure 32: the receipt file saved

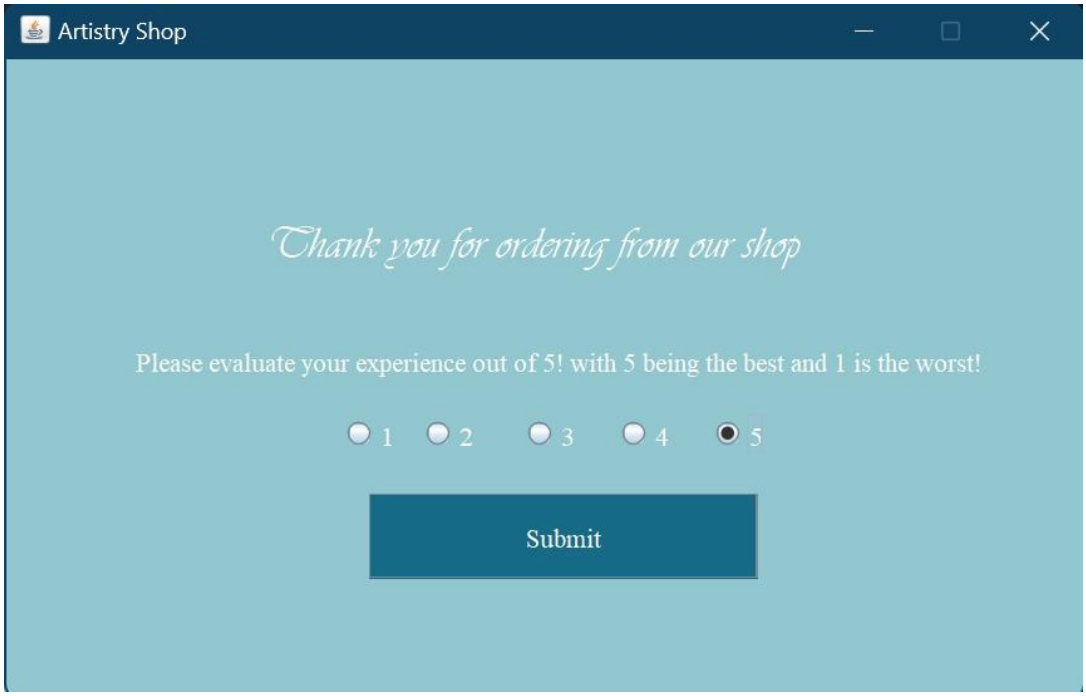


Figure 33: evaluate